

Farzaneh Fayazbakhsh

Spatial Transcriptomics & Bioinformatics Researcher | Cancer Immunology

📍 Stockholm, Sweden ✉ fa.fayazbakhsh@gmail.com ☎ 079-346 08 88 🌐 fari-labwitch

Personal Statement

Interdisciplinary bioinformatics and molecular-biology researcher with direct experience in single-cell and spatial transcriptomics, tumor models and sequencing-data analysis. I use Python, R, Linux and reproducible workflows to translate complex biological questions into rigorous, collaborative analyses, and I am eager to contribute to data-driven spatial immune-niche research in cancer.

Bioinformatics & Computational Biology Projects

Automated Raw Sequencing Data Quality Control Pipeline

- Built an automated, parallelized Bash orchestration workflow designed for execution on High-Performance Computing (HPC) clusters via SLURM scheduling.
- Deployed FastQC and fastp dynamically inside isolated, version-controlled Singularity and Apptainer containers to automate raw sequence pairing, quality checking, and adapter trimming.
- Implemented multi-core processor allocation inside container runtimes to optimize throughput when executing string substitutions and batch file calculations.
- Orchestrated MultiQC container tasks to aggregate and output comprehensive data quality matrices, tracking processing adjustments across all sample cohorts simultaneously.

Tumor Microenvironment Profiling: scRNA-Seq & Spatial Transcriptomics

- Engineered end-to-end transcriptomic pipelines in R using Seurat and SCTransform to map high-dimensional gene expression across breast cancer tissue cohorts.
- Executed data pre-processing, Quality Control (QC), linear (PCA), and non-linear (t-SNE / UMAP) dimensionality reduction alongside graph-based clustering to resolve distinct cell populations and structural tissue features.
- Containerized both analysis environments using Docker and Docker Compose, ensuring full cross-platform compatibility and seamless runtime execution across varied infrastructure platforms.
- Managed pipeline documentation, software repositories, and strict workflow reproducibility utilizing Git and GitHub version control standards.

Computational Biology & Bioinformatics Skills

Programming & Pipeline Automation: Python, R, UNIX/Linux Shell, Nextflow (pipeline development), Git, and GitHub for version control

Genomics, High-Throughput Sequencing & Genetics: Practical experience in NGS data analysis, Whole Genome Sequencing (WGS), RNA-Seq processing, GWAS, QTL analysis, gene mapping, structural variation, metagenomics, genome annotation, and phylogenomics (MEGA, Clustal Omega)

Structural Biology, Modeling & Machine Learning: Sequence analysis, primer design, AlphaFold (protein structure prediction), PyMOL (molecular visualization), and foundational familiarity with deep learning frameworks

Infrastructure, HPC & DevOps: Hands-on experience with High-Performance Computing (HPC) clusters, containerization (Docker, Docker Compose, Apptainer/Singularity), and implementation of FAIR Research Data Principles

Research Experience

Karolinska Institute & SciLifeLab, Master's Thesis Research

2026–present

- Investigating replication-stress-induced genomic instability at common fragile sites using targeted long-read sequencing.
- Performing neuroblastoma cell culture, drug treatments, DNA extraction, and long-range PCR.
- Developed bioinformatics workflows using Python, Linux and HPC environments for long-read sequencing analysis.
- Performed structural variant discovery, breakpoint characterization and endogenous barcode identification.

KTH Royal Institute of Technology & SciLifeLab , Research Engineer Sweden - Independently developed Tumor-on-a-Chip models for in vitro anti-cancer drug screening. - Optimized layer-by-layer cellulose nanofibril coatings to enhance cancer spheroid formation. - Contributed to projects on Brain-on-a-Chip models for metabolic studies. - Functionalized electrochemical sensors for ketone detection.	2023–2024
Tehran University of Medical Sciences , Master's Thesis Research Iran - Synthesized gold nanoparticles conjugated with polyphenols. - Developed microfluidic devices for endothelial cell culture. - Modeled early-stage endothelial dysfunction and conducted oxidative stress assays. - Assisted in preparing a PET/PDMS/Collagen nanofibrous membrane for microfluidic cell culture.	2018–2022
Gen-Iran Lab , Pathology and Cytology Intern Tehran, Iran - Hands-on experience in tissue preparation, fixation, embedding, and sectioning. - Performed histological and immunohistochemical staining.	2023

Wet Lab & Experimental Expertise

Cell Culture & Cancer Biology: 2D & 3D cell culture, tumor spheroid formation and organoid culture, in vitro cancer biology drug screening

Molecular Biology & Genomics Workflows: DNA extraction, downstream cellular and molecular assessments, PCR / Real-Time PCR, agarose gel electrophoresis, Agilent TapeStation analysis, flow cytometry, fluorescent microscopy

Microfluidics & Biosensors: Organ-on-a-Chip modeling, PDMS-based microfluidic systems fabrication, surface and electrochemical sensor functionalization

Histology & Pathology: Histological sample preparation, H&E staining, immunohistochemistry (IHC)

Nanotechnology & Nanomaterial Synthesis: Green synthesis of gold nanoparticles, nanofiber electrospinning, cellulose nanofibrils (CNF) coating optimization

Peer-Reviewed Publications

Evaluating the Antioxidant Potential of Resveratrol-Gold Nanoparticles in Preventing Oxidative Stress in Endothelium on a Chip Farzaneh Fayazbakhsh , Fatemeh Hataminia, Houra Mobaleghol Eslam, Mohammad Ajoudanian, Sharmin Kharrazi, Kazem Sharifi, Hossein Ghanbari (Scientific Reports)	2023
Metabolic Assessment of iPSC-Derived Neurons Under Ketogenic and Normal Diets: Ketone Sensor Development and Comparative Analysis of Neuronal Metabolism * Both authors contributed equally to this work (co-first authors). Rohollah Nasiri*, Farzaneh Fayazbakhsh *, Reyhaneh Sanei, Tingting Wu, Nayere Taebnia, Rouhollah Habibey, Omid Fotouhi, John Coggnetti, Volker M. Lauschke, Aman Russom, Anna Herland (CellPress)	2026
Assessing Layer-by-Layer Assembly of Cellulose Nanofibrils in Pancreatic Tumor Spheroid Formation Negar Abbasi Aval, Ekeram Lahchaichi, Oana Tudoran, Farzaneh Fayazbakhsh , Rainer Heuchel, Matthias Löhr, Torbjörn Pettersson, Aman Russom (Biomedicines)	2023
Characterization and Numerical Simulation of a New Microfluidic Device for Cell Investigations H Meslam, F Hataminia, H Asadi-Saghandi, F Fayazbakhsh , N Tabatabaei, Hossein Ghanbari (Frontiers in Lab on a Chip Technologies)	2024
Nanofibers in Respiratory Masks: An Alternative to Prevent Pathogen Transmission Tabatabaei SN, Faridi-Majidi R, Boroumand S, Norouz F, Rahmani M, Rezaie F, Fayazbakhsh F , Faridi-Majidi R (IEEE Transactions on NanoBioscience)	2022

Education

- M.Sc. Swedish University of Agricultural Sciences**, Bioinformatics 2025–present
- Highest grade (5/5) in Bioinformatics course
 - Thesis: Exploring the Potential of Fragile Sites as Markers for Lineage Tracing
- KTH Royal Institute of Technology**, Single-Cell & Spatial Transcriptomics Data Analysis Course 2024
- M.Sc. Tehran University of Medical Sciences**, Medical Nanotechnology 2017–2022
Iran
- Awarded as the Top Master's Graduate (GPA: 4/4)
 - Thesis: Evaluating the Antioxidant Potential of Resveratrol-Gold Nanoparticles in Preventing Oxidative Stress in Endothelium on a Chip
- B.Sc. Alzahra University**, Cellular and Molecular Biology (Biotechnology Specialization) 2012–2016
Iran

Conference Presentations

- Spheroid Formation in Cellulose Nanofibrils-Coated Microfluidic Chips** 2024
µTAS 2024, Montreal, Canada
Poster presentation (first author)
- Design of a Microfluidic Electrochemical Ketone Sensor** 2023
Swedish Microfluidics in Life Science, KTH, Sweden
Oral presentation (co-author)
- Investigation of Effect of Different Diets on Brain Energy Metabolism Using Organ-on-a-Chip** 2023
MPS World Summit, Berlin, Germany
Poster presentation (co-author)
- Preparation of PET/PDMS/Collagen Nanofibrous Membrane Embedded in a Microfluidic Device for Organ-on-Chip Applications** 2022
The 3rd Conference on Nanofibers, Tehran University of Medical Sciences
Poster presentation (co-author)

Professional & Personal Skills

- Bridging Wet & Dry Lab:** Confident working in both experimental labs and computational environments, making it easy to translate biological questions into data-driven answers.
- Independent Problem Solver:** Highly autonomous in research environments, with a proven ability to troubleshoot bottlenecks and quickly teach myself new tools to keep projects moving forward.
- Collaborative Team Player:** Experienced in international, interdisciplinary teams such as KTH, Karolinska Institute and SciLifeLab, with a strong focus on knowledge-sharing and helping colleagues.
- Clear Communicator & Fast Learner:** Skilled at explaining complex data clearly through peer-reviewed publications and conference presentations; highly adaptable and quick to pick up new scientific topics and skills.

Teaching & Mentorship

- KTH Royal Institute of Technology**, Intern Supervisor
Sweden
- Guided an intern in laboratory techniques.
- Volunteer Service**, Science Teacher
- Taught science to underprivileged children.

Awards & Honors

- Top Master's Graduate** 2023
Tehran University of Medical Sciences
- LUSH Prize Candidate** 2022
Shortlisted for the Young Researcher Award.

Darolfonoon Student Award For voluntary assistance in nanofiber mask production during the COVID-19 pandemic.	2020
Top Article Award First National and Student Conference on Applied Biology, Alzahra University, Iran Prize winner.	2014
Top 5% in Iran's Konkour for Master's Degree University Entrance Iran Ranked 37th in Iran's national Konkour for the Master's Degree University Entrance Exam.	2016

References

Dr. Kasper Karlsson

Department of Oncology-Pathology, Karolinska Institute
kasper.karlsson@ki.se

Dr. Erik Bongcam Rudloff

Department of Animal Biosciences, Swedish University of Agricultural Sciences
erik.bongcam@slu.se

Dr. Aman Russom

Nanobiotechnology Division, Protein Science Department, KTH Royal Institute of Technology
aman.russom@scilifelab.se

Cheng-de (Ernest) Liu

Department of Oncology-Pathology, Karolinska Institute
chengde.liu@ki.se

Dr. Anna Herland

Nanobiotechnology Division, Protein Science Department, KTH Royal Institute of Technology
aherland@kth.se

Dr. Rohollah Nasiri

Radiation Oncology Department, Stanford University
rhnasiri@stanford.edu